

Chapter 5

Saturday, November 09, 2024

12:40 PM

Topics:

- The characteristic equation

The Characteristic Equation

$$\begin{bmatrix} 10228.5 & 2880 \\ 2880 & 1307.34 \end{bmatrix} A \times I\lambda = 0$$

$$(10228.5 - \lambda)(1307.34 - \lambda) - (2880)^2$$

$$-10228.5\lambda + \lambda^2 + 13372127.19 - 1307.34\lambda - 8294400$$

$$\lambda^2 - 11535.84\lambda + 5077727.19 = 0$$

$$\frac{-(-11535.84) \pm \sqrt{(-11535.84)^2 - 4(1)(5077727.19)}}{2}$$

$$\frac{11535.84 \pm 10619.07226}{2} \quad \begin{matrix} -\circ 458.38387 \\ +\circ 11077.45613 \end{matrix}$$

- The basic power method

The Basic Power Method (largest eigenvalue)

start with vector $x = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$$A = \begin{bmatrix} 4 & 2 & -2 \\ -2 & 8 & 1 \\ 2 & 4 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 & -2 \\ -2 & 8 & 1 \\ 2 & 4 & -4 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \quad \begin{matrix} (4 \times 1) + (2 \times 1) + (-2 \times 1) = 4 \\ (-2 \times 1) + (8 \times 1) + (1 \times 1) = 7 \\ (2 \times 1) + (4 \times 1) + (-4 \times 1) = 2 \end{matrix} \quad \begin{bmatrix} 4 \\ 7 \\ 2 \end{bmatrix}$$

$$\begin{matrix} (4 \times \frac{4}{7}) + (2 \times 1) + (-2 \times \frac{2}{7}) = \frac{26}{7} = 3.714 \\ (-2 \times \frac{4}{7}) + (8 \times 1) + (1 \times \frac{2}{7}) = \frac{58}{7} = 8.286 \\ (2 \times \frac{4}{7}) + (4 \times 1) + (-4 \times \frac{2}{7}) = 4 \end{matrix} \quad \begin{bmatrix} 3.714 \\ 8.286 \\ 4 \end{bmatrix}$$

largest eigenvalue will converge

- The inverse power method

Inverse Power Method (smallest eigenvalue)

$A^{-1} \begin{bmatrix} 0.333 & 0 & -0.166 \\ 0.0555 & 0.111 & 0 \\ 0.222 & 0.111 & -0.333 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$ Inverse of matrix A

$$\begin{aligned} (0.333 \times 1) + (0 \times 1) + (-0.166 \times 1) &= 0.1664 \\ (0.0555 \times 1) + (0.111 \times 1) + 0 &= 0.1665 \\ (0.222 \times 1) + (0.111 \times 1) + (-0.333 \times 1) &= 0 \end{aligned}$$

$\begin{bmatrix} 0.9993 \\ 1 \\ 0 \end{bmatrix}$ Smallest eigenvalue will converge

$$\begin{aligned} (0.333 \times 0.9993) + (0 \times 1) &= 0.3327 \\ (0.0555 \times 0.9993) + (0.111 \times 1) &= 0.166 \\ (0.222 \times 0.9993) + (0.111 \times 0) &= 0.2218 \end{aligned}$$

$\begin{bmatrix} 1 \\ 0.5 \\ 0.668 \end{bmatrix}$

- The shifted power method

Shifted Power Method

Start with vector $\begin{bmatrix} 0 & 1 & 1 \end{bmatrix}$ and $\lambda = 7.6898$

$\begin{bmatrix} 4 & 2 & -2 \\ -2 & 8 & 1 \\ 2 & 4 & -4 \end{bmatrix} \quad [x]_{i+1} = [A - \lambda I][x]_i$

STARTING WITH LARGEST EIGENVALUE
CALCULATES SMALLEST EIGENVALUE & VICE VERSA

$$[x]_2 = \begin{bmatrix} 4 & 2 & -2 \\ -2 & 8 & 1 \\ 2 & 4 & -4 \end{bmatrix} - 7.6898 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 4-7.6898 & 2 & -2 \\ -2 & 8-7.6898 & 1 \\ 2 & 4 & -4-7.6898 \end{bmatrix} = \begin{bmatrix} -3.6898 & 2 & -2 \\ -2 & 0.3102 & 1 \\ 2 & 4 & -11.6898 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} (-3.6898 \times 0) + 2 \times 1 + (-2 \times 1) &= -2 \\ (-2 \times 0) + (0.3102 \times 1) + (1 \times 1) &= 1 \\ (2 \times 0) + (4 \times 1) + (-11.6898 \times 1) &= -11.6898 \end{aligned}$$

$\begin{bmatrix} -2 \\ 1 \\ -11.6898 \end{bmatrix}$ LARGEST ABSOLUTE VALUE

$[x]_2 = \begin{bmatrix} 0.1711 & -0.0855 & 1 \end{bmatrix}$

- QR Factorization theory

QR Factorisation

- $[a]$ and $[b]$ are similar if $[a] = [c]^{-1}[b][c]$ where c is an invertible matrix. $[a]$ and $[b]$ have the same eigenvalues
- $[a]$ is orthogonal if $[a]^{-1} = [a]^T$
- $[a]$ is symmetric if $[a] = [a]^T$
- The eigenvalues of an upper triangular matrix are its diagonal elements

let $[a]_1 = [Q]_1[R]_1$ where $[Q]_1$ is orthogonal & $[R]_1$ is upper triangular

$$[a]_2 = [R]_1[Q]_1$$

$\therefore [a]_2 = [Q]_1[a]_1[Q]_1$ $[a]_1$ & $[a]_2$ are similar

$$[a]_3 = [R]_2[Q]_2$$

$\therefore [a]_3 = [Q]_2[a]_2[Q]_2$ $[a]_2$ & $[a]_3$ are similar

Iterate until $[a]_{n-1}$ results an upper triangular matrix whose eigenvalues are its diagonal elements

The householder matrix is both symmetric & orthogonal

$$[H] = [H]^T = [H]^{-1} \therefore [a] = [H][b][H] \therefore b \text{ similar to } a$$

QR Factorization example

QR factorisation

$$a = \begin{bmatrix} 6 & -7 & 2 \\ 4 & -5 & 2 \\ 1 & -1 & 1 \end{bmatrix} \quad [c] = \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix} \quad [e] = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\|c\|_2 = \sqrt{36+16+1} = 7.2801$$

$$[v] = [c] + \|c\|_2 [e] = \begin{bmatrix} 6 & 4 & 1 \\ 0 & 0 & 0 \end{bmatrix} + 7.2801 \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 13.2801 & 4 & 1 \end{bmatrix}$$

$$[v][v]^T = \begin{bmatrix} 13.2801 & 4 & 1 \\ 0 & 0 & 0 \end{bmatrix} = 176.3611$$

$$[v][v]^T = \begin{bmatrix} 13.2801 & 4 & 1 \\ 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 176.3611 & 53.1204 & 13.2801 \\ 53.1204 & 16 & 4 \\ 13.2801 & 4 & 1 \end{bmatrix}$$

$$[H] = [I] - \frac{2}{[v][v]^T} [v][v]^T$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{176.3611} \begin{bmatrix} 176.3611 & 53.1204 & 13.2801 \\ 53.1204 & 16 & 4 \\ 13.2801 & 4 & 1 \end{bmatrix} = \begin{bmatrix} -0.8242 & -0.5494 & -0.1374 \\ -0.5494 & 0.8345 & -0.0414 \\ -0.1374 & -0.0414 & 0.9897 \end{bmatrix}$$

Once the householder matrix is constructed $[a]$ can be factored into $[Q]^T [R]^T$

$$[Q]^T = [H]^T = \begin{bmatrix} -0.8242 & -0.5494 & -0.1374 \\ -0.5494 & 0.8345 & -0.0414 \\ -0.1374 & -0.0414 & 0.9897 \end{bmatrix}$$

$$[R]^T = [H]^T [a] = \begin{bmatrix} 7.2801 & 8.6537 & -2.8846 \\ 0 & -0.2851 & 0.5288 \\ 0 & 0.1787 & 0.6322 \end{bmatrix}$$

$$[c] = \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix} \quad [e] = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad \|c\|_2 = \sqrt{0^2 + (-0.2851)^2 + 0.1787^2} = 0.3365$$

$$[v] = [c] + \|c\|_2 [e] = \begin{bmatrix} 6 \\ -0.2851 \\ 0.1787 \end{bmatrix} + 0.3365 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 6.3365 \\ -0.2851 \\ 0.1787 \end{bmatrix}$$

QR Factorization

$$[V][V]^T = \begin{bmatrix} 0 & -0.6215 & 0.1787 \\ 0.1787 & 0.6215 & 0 \end{bmatrix} = 0.4183$$

$$[V][V]^T = \begin{bmatrix} 0 & -0.6215 & 0.1787 \\ 0.1787 & 0.6215 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.3864 & -0.1111 \\ 0 & 0.1111 & 0.0319 \end{bmatrix}$$

$$[H]^2 = [I] - \frac{2}{[V][V]^T} [V][V]^T = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \frac{2}{0.4183} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0.3864 & -0.1111 \\ 0 & 0.1111 & 0.0319 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.8474 & 0.5311 \\ 0 & 0.5311 & 0.8473 \end{bmatrix}$$

$$[Q]^2 = [Q]^T [H]^2 = \begin{bmatrix} -0.8242 & -0.5494 & -0.1374 \\ -0.5494 & 0.8345 & -0.4114 \\ -0.1374 & -0.4114 & 0.9877 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.8474 & 0.5311 \\ 0 & 0.5311 & 0.8473 \end{bmatrix}$$

$$= \begin{bmatrix} -0.8242 & 0.3927 & -0.4082 \\ -0.5494 & -0.4291 & 0.4082 \\ -0.1374 & 0.5607 & 0.8166 \end{bmatrix}$$

$$[R]^2 = [H]^2 [R] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -0.8474 & 0.5311 \\ 0 & 0.5311 & 0.8474 \end{bmatrix} \begin{bmatrix} -7.2801 & 8.6537 & -2.8846 \\ 0 & -0.2851 & 0.5288 \\ 0 & 0.1787 & 0.6322 \end{bmatrix}$$

$$= \begin{bmatrix} -7.2801 & 8.6537 & -2.8846 \\ 0 & 0.3365 & -0.1123 \\ 0 & 0 & 0.8165 \end{bmatrix} \Rightarrow \text{upper triangular} \\ \Rightarrow \text{eigenvalues}$$

$$[a] = [Q]^2 [R]^2$$